

peak-power analyses.”

Absorptive RF power meters. These utilise a power sensor to generate a signal proportional to the power level entering the sensor. Sensor signal is coupled with the main instrument to process the results and display the readings.

These meters measure total input power, regardless of frequency. These have digital readouts and significant levels of digital signal processing. This enables a variety of signal types to be measured.

Through-line RF power meters. These meter take a sample of the power flowing along a feed line and use it to indicate power level. These are used on live systems, such as radio transmitters, as a check for outgoing power. These meters can be used to check the power travelling in either direction. Total input power is measured regardless of frequency.

Spectrum analysers. These instruments can measure RF power levels on a particular frequency but not the total power entering on all frequencies. These make measurements of signals in the frequency domain.

Types of RF power sensors

Maximum input is the signal power level that can be fed to the power sensor without damaging it. If signal level beyond maximum input is fed, the sensor may get permanently damaged or become non-linear. Frequency range of measurement and sensitivity depend on the power sensor used. A single sensor may not cover the entire range of frequencies or power levels.

For higher power-level measurements and very-low power-level measurements, separate sensors may be required. So, it is advisable to select one depending on application requirements.

Main factors for consideration while selecting a power sensor are frequency, signal and power range to be measured, and connector type.

There are two types of sensors used for RF power measurement, as given below.

Thermal/heat-based sensors. These measure power purely based on

FEATURES OF LATEST POWER METERS

Instrument	Specifications
R&S NRP2 power meter by Rohde & Schwarz	• -67dBm to +45dBm level range (sensor dependent) • 9kHz to 110GHz frequency range (sensor dependent) • 1500 measurements per second in buffered mode • 1.5% linearity uncertainty
E4 416A/E4 417A EPM-P series power meters by Keysight Technologies	• -70dBm to +44dBm average power range (sensor dependent) • 1000 readings per sec • 9kHz to 110GHz frequency range (sensor dependent) • $\pm 1.7\%$ linear absolute accuracy
ML2430A CW power meter by Anritsu	• 100kHz to 65GHz (sensor dependent) • -70 to +20dBm dynamic range (sensor dependent) • More than 600 readings per second (per input channel) • $\pm 1.2\%$ power accuracy

heat dissipation, and can be used to measure any waveform, including CW, AM, FM, PM and pulsed. Thermal sensors are further divided into two types:

Thermistor RF power sensor. This sensor works on the fact that temperature in an RF load increases due to incident input RF signal.

Thermocouple power sensor. This is based on the thermocouple principle that says, at different temperatures, small potential is induced at metal junctions. It can be used to measure average power levels.

Diode-based power sensors. These can measure signals at very low levels (below -70dBm). These have a wider dynamic range, but their non-linear characteristics need compensation to achieve accurate average and peak power measurements.

These sensors can measure the average power of CW signals and peak power of modulated signals. Schottky barrier diodes are often used as RF power sensors, as these have a much smaller level of stored charge and a low forward conduction turn on point.

Criteria for selecting the right equipment

Different types of power sensors are suited for different types of RF power measurements. Peak power of a signal is not the same as its average power. This is because of the phase relationship between tones. Some types of RF power sensors support average power measurement, whereas others support pulse power or peak-envelope power measurements.

Take care to select a power sensor that covers the required frequency range. When there is a need for extra frequency ranges, a different power

sensor can be used along with the same basic power meter. If there is a need for extended frequency range in the next five years, it is better to include those frequencies now instead of purchasing additional instruments after five years. An instrument that can be augmented easily for future requirements by installing plug-in modules can also be chosen.

The most important and essential factor in selecting test instruments is their reliability, accuracy, sensitivity and cost.

The service office should be in close vicinity of the setup.

If the instrument is procured from abroad, the manufacturer should have local service centres.

Specifications should take care of current requirements as well as possible future requirements to some extent.

Application areas

Taking measurements requires an informed choice of technique and equipment. Development of radio systems requires consistent measurements of RF or microwave power levels. There are many other applications that need to measure output power from a power amplifier, or received power to the system, as given below.

- Measuring output power of IC amplifier
- Verifying compliance to ensure public safety and that RF signals do not overpower devices on adjacent frequencies
- Measuring signals between devices

To sum up

The key determinant of performance of an RF system is its output power. This requires measurement of signal power at every stage, from design to prototyping.

Calibration can provide greater assurance of RF instrument accuracy. ISO 9001-compliant services have been conducted within ISO 17025-accredited laboratories to check RF power sensor testing capabilities. **EFY**